

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 – SUB-STRAND 3.3 INTRODUCTION TO ELECTRONICS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: Why does a silicon solar panel on a Kajiado farm produce less power in the hot afternoon sun (about 35 °C) than on a cool morning (about 18 °C), and what is so special about silicon that it can act as both a conductor and an insulator depending on the conditions? Use everything you learned in Lessons 1 to 7: your circuit tests, band diagrams, doping diagrams, the PhET simulation and the case studies. Answer each section in the blank space under its prompt. Write in full sentences. Use the scientific words we learned (valence band, conduction band, band gap, electron-hole pair, doping, n-type, p-type, p-n junction). Where a prompt asks for a diagram, draw it neatly and label it. Where a prompt asks for evidence, refer to specific results from our lessons.

Part 1: What our circuit tests showed

In Lesson 1 you tested materials in a 1.5 V circuit with an LED. Here are the results:

Material	LED brightness
Copper wire	Fully bright
Steel nail	Fully bright
Plastic sufuria handle	Off
Rubber eraser	Off
Graphite pencil lead	Dim
Silicon chip	Dim

(a) Use the table to sort the materials into conductors, insulators and materials that conduct only partially, and give your reason for each group. (b) Name the category of materials that conduct only partially, and explain why a material in that category has its name. (c) Why could your earlier ideas about simple

circuits and Ohm's Law not explain the silicon result or the Kajiado solar panel?

Write your answer here:

Part 2: The energy band model and silicon's dual identity

(a) Draw and label simple energy band diagrams for a metal such as copper, an insulator such as the plastic handle or rubber, and a semiconductor such as silicon. Show the valence band, the conduction band and the band gap on each. (b) Explain how the size of the band gap decides whether a material is a conductor, an insulator or a semiconductor. (c) Silicon's band gap is about 1.1 eV. Use this to explain why silicon can behave like an insulator in darkness but conduct when sunlight falls on it. Then explain what happens to the number of electron-hole pairs in pure silicon when it is heated, and why this is a different question from how well a solar panel performs.

Write your answer here:

Part 3: Doping and the p-n junction

The silicon in a real solar panel is not pure. (a) What is doping, and why do engineers do it? (b) Explain how phosphorus makes n-type silicon and how boron makes p-type silicon. Say which charge carriers are the majority in each. (c) Describe what a p-n junction does to the flow of charge, using what you saw in the PhET simulation about forward bias and reverse bias. (d) Explain why the p-n junction matters for the Kajiado solar panel.

Write your answer here:

Part 4: Solving the Kajiado solar panel mystery

Mama Akinyi's silicon panel in Kajiado gives less power at about 35 °C on a hot afternoon than at about 18 °C on a cool morning, even though the afternoon sun is strong. Write a Claim–Evidence–Reasoning explanation. (a) Claim: answer the driving question in one or two sentences. (b) Evidence: give at least two pieces of evidence from our lessons (for example the temperature data, the band diagrams, the PhET simulation or the case study graph). Say what each piece shows and be careful about how closely its temperature range matches the Kajiado panel. (c) Reasoning: use the energy band model, the p-n junction and electron-hole pairs to explain how a higher temperature leads to your claim. (d) Explain why the claim 'more heat means more electricity' does not hold for this panel, and why it matters that silicon is neither copper nor rubber.

Write your answer here:

Part 5: Semiconductors in Kenyan life

(a) Name two Kenyan technologies, other than the Kajiado panel, that depend on semiconductors. For each one, say which semiconductor idea from our lessons it uses (band gap, doping, p-n junction or temperature effects). (b) Explain how understanding and engineering semiconductor materials has changed everyday life in Kenya. (c) In which way did your thinking change between Lesson 1 and now?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of the energy band model	Correctly describes valence band, conduction band and band gap for a metal, an insulator and silicon (about 1.1 eV), and uses them to explain why silicon conducts in light but not in darkness.	Describes the bands and gaps for most materials and explains silicon's conduction, with small gaps in detail or vocabulary.	Mentions bands or gaps but confuses which material has which gap, or does not link the gap to conduction.
Explanation of doping and the p-n junction	Accurately explains n-type (phosphorus, electron donor) and p-type (boron, holes), the majority carriers, and how the junction gives one-way current, linked to the solar panel.	Explains n-type and p-type correctly and mentions one-way flow, but the link to the panel is brief.	Confuses n-type and p-type or the carriers, or does not explain the junction's role.
Claim–Evidence–Reasoning on the Kajiado phenomenon	Gives a clear claim, at least two specific pieces of evidence from the lessons (noting how well each matches the Kajiado temperatures), and logical reasoning from higher temperature (35 °C vs 18 °C) through lattice vibrations and electron-hole recombination to reduced power output.	Gives a clear claim with some evidence and mostly correct reasoning, with a gap in the chain of cause and effect.	States that it is hotter and gives less power without a mechanism, or says more heat gives more electricity.
Application to Kenyan contexts	Gives two accurate Kenyan examples	Gives examples with a general link to	Gives examples without linking them to

and scientific communication	linked to specific semiconductor ideas, reflects on the change in thinking since Lesson 1, and uses scientific terms correctly, with clear labelled diagrams.	semiconductors and uses most terms correctly; diagrams are mostly labelled.	the physics, uses few scientific terms, or leaves diagrams unlabelled.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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